ACECORE

User Manual

A COMPLETE GUIDE TO TRUSTING YOUR POWERCELL

ACECORE

We Kept This Page Specially For You

You're Amazing!

Dear Beautiful Human,

Thank You For Believing In Acecore And Choosing To Be Part Of Our Journey. Your Trust Means More To Us Than You May Ever Know.

Every Time We See A Powercell Out In The World, We're Reminded Of The People Who Believed In The Vision Early. And You Will Always Be One Of Those People.

Thank You For Trusting Us, Supporting Us, And Giving Us The Opportunity To Serve You.

With Love,

Caleb Iwuoha (CEO)

Table Of Contents

A guide built to be read in any order

POWERCELL SAFETY INSTRUCTIONS

PART ONE - ABOUT YOUR POWERCELL

Meet Your Powercell 07

Powercell Status 08

Power State 09

Smart Socket 10

Energy Flow 11

PART TWO — INSTALLATION

Installation Safety 13

Installation Rules 14

Quick Setup 15

PART THREE — THE ACECORE APP

Getting Started 16

Your Dashboard 19

Insights & Recommendations 20

Remote Control 21

Troubleshooting 22

Account & Device Support 23

FAQs 24

Privacy & Security 25

Contact Support 26

PART FOUR — LIVING WITH POWERCELL

Everyday Care 27

Ongoing Safety 28

Specifications 29

Support & Warranty 30

Powercell Safety Instructions

SAVE THESE IMPORTANT SAFETY INSTRUCTIONS

Powercell installation and service require knowledge of high voltage electricity and should only be performed by Acecore Certified Installers. Acecore assumes no liability for injury or property damage due to repairs attempted by unqualified individuals or a failure to properly follow these instructions. These warnings and cautions must be followed when using Powercell.

Symbols in This Document

This manual uses the following symbols to highlight important information:

WARNING: indicates a hazardous situation which, if not avoided, could result in injury or death.

CAUTION: indicates a hazardous situation which, if not avoided, could result in minor injury or damage to the equipment.

NOTE: indicates an important step or tip that leads to best results, but is not safety or damage related.

General Information

WARNING: Read this entire document before installing or using Powercell. Failure to do so or to follow any of the instructions or warnings in this document can result in electrical shock, serious injury, or death, or can damage Powercell, potentially rendering it inoperable.

WARNING: Electric shock hazard. The DC conductors of any connected PV system are normally ungrounded but will become intermittently grounded without indication when the inverter measures the PV array isolation. Disconnect all sources of supply before servicing.

WARNING: A battery can present a risk of electrical shock, fire, or explosion from vented gases. Observe proper precautions.

WARNING: This product can expose you to chemicals including lithium compounds. These chemicals are known to cause cancer and birth defects or other reproductive harm. For more information, visit ***www.acecore.com*** You cannot be exposed to these chemicals without opening the external casing. Only members of the Acecore Service team and other Acecore certified technicians should open the external casing.

WARNING: Powercell installation must be carried out only by Acecore Certified Installers who are licensed electricians.

WARNING: Powercell is heavy. Use of lift equipment is recommended.

WARNING: Use Powercell only as directed.

Powercell Safety Instructions

WARNING: Do not attempt to open, disassemble, repair, tamper with, or modify Powercell. Powercell and its components are not user serviceable. Batteries in Powercell are not replaceable. Contact Acecore Support or the Certified Installer who installed the system for any repairs.

WARNING: Do not insert foreign objects into any part of Powercell.

WARNING: Do not expose Powercell or its components to direct flame.

WARNING: Do not install Powercell near heating equipment.

WARNING: Do not immerse Powercell or its components in water or other fluids.

CAUTION: Powercell is not designed nor warrantied for non-stationary applications.

CAUTION: Risk of electric shock. Do not remove cover, there are no user serviceable parts inside. Refer servicing to qualified personnel.

CAUTION: Both AC and DC voltage sources may be terminated inside this equipment. Each circuit must be individually disconnected before servicing.

CAUTION: Do not use cleaning solvents to clean Powercell, or expose Powercell to flammable or harsh chemicals or vapors.

CAUTION: Do not use fluids, parts, or accessories other than those specified in this manual, including use of non-genuine Acecore parts or accessories, or parts or accessories not purchased directly from Acecore

CAUTION: Do not place Powercell in a storage condition for more than one (1) month, or permit the electrical feed on the Powercell to be severed for more than one (1) month. Please reach out to your installer for more information if needed.

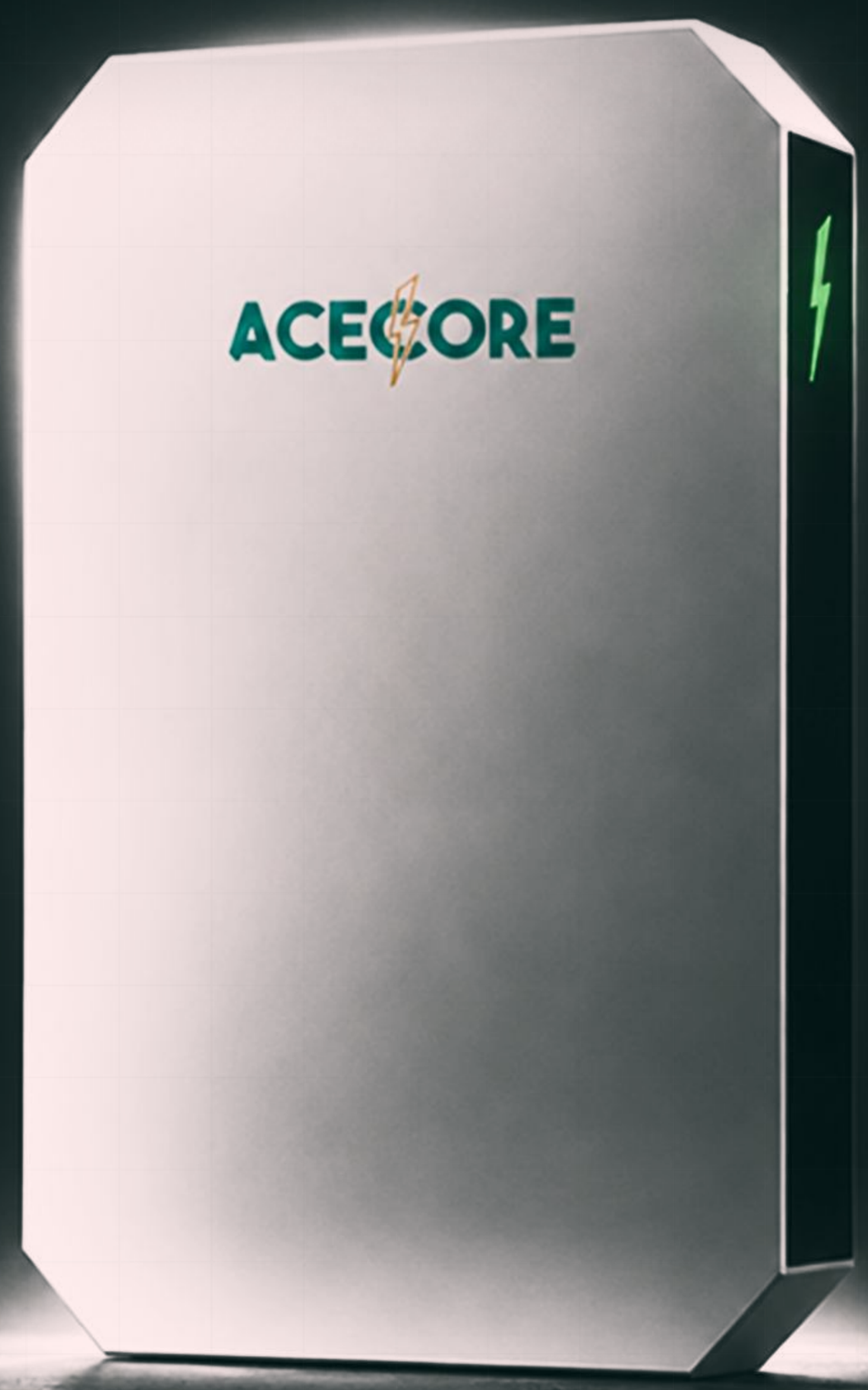
CAUTION: Do not paint any part of Powercell, including any internal or external components such as the exterior shell or casing.

ACECORE

Part One

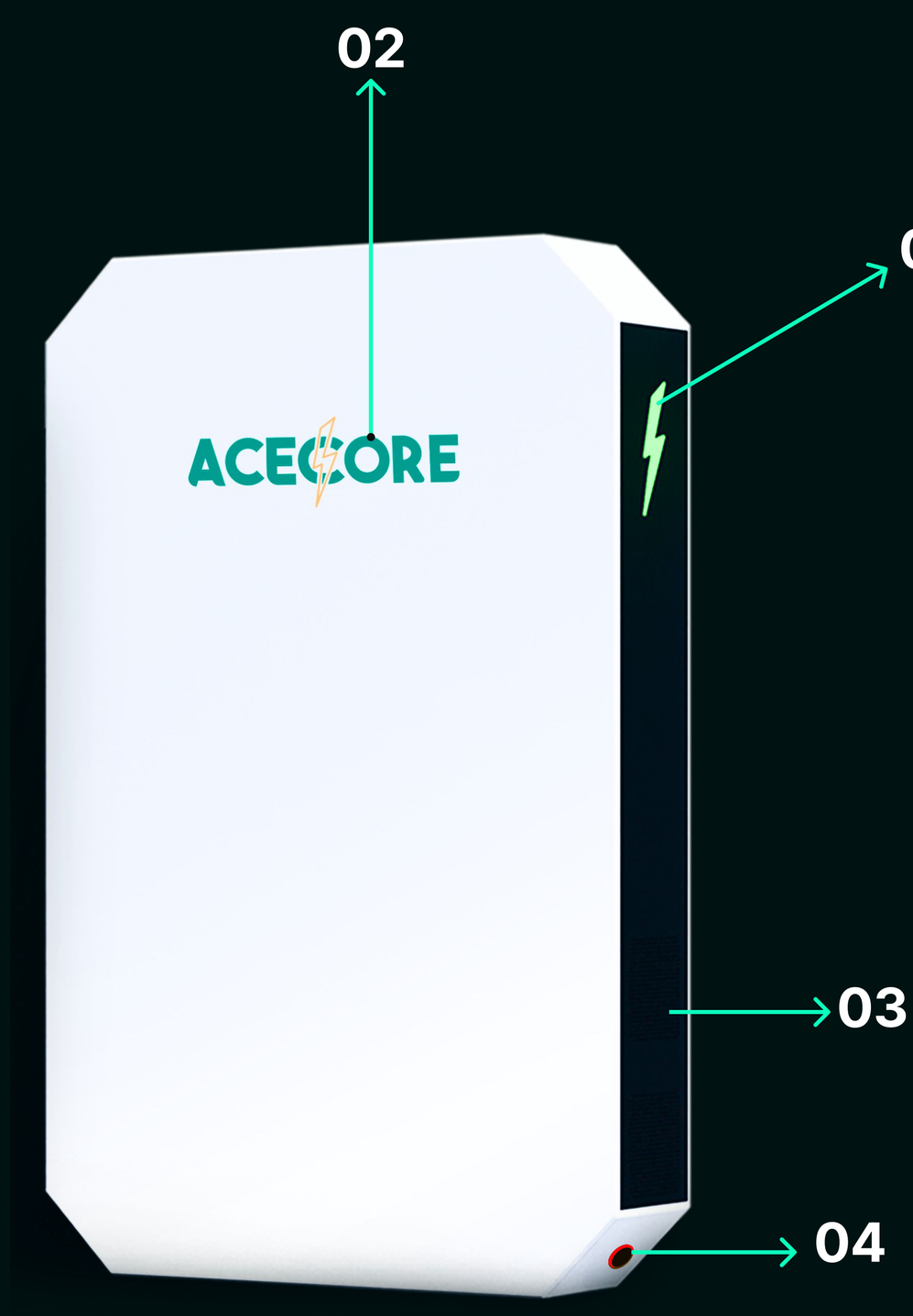
The Powercell

Contains everything you need to properly
use, care for and enjoy your Powercell



Get to know your PowerCell

One tower. Four things to recognise. Once you know these, you know everything the unit is telling you.



01 Status Light

The lightning-bolt window: It shows your charge level while you charge, and warns you the moment your battery runs low. See page 4.

02 Acecore Panel

The illuminated nameplate on the front means your powercell is turned ON whenever its on and tells you if your powercell is turned OFF if its not lightened on.

03 Vents

Draws in cool air to keep the system running efficiently. Keep it free of dust, cloth, and clutter at all times. On the other side of the system it Releases warm air from inside the unit. Leave clear space in front of & around the vents, never box the system in.

04 System Control Button

The button on the lower right is used for powering ON the system for the first time. It doesn’t switch ON the inverter, it only powers ON the electronics. When the button is ON, you would see the broadcast of the Powercell setup for the App.
It also disconnects the battery entirely from the system (Hold till it beeps)

One light. Everything you need to know.

The lightning bolt tells you whether your system is fully charged or very low



0% - 39%
Critically Low

This indicates that the battery is critically Low.

RED · FLICKERING



40% - 69%
Mid Level

Powercell charge is at Mid level.

ORANGE · FLICKERING



70% - 100%
Fully Charged

The Powercell is topped up and the system is running normally.
No action needed.

GREEN · STEADY



Charging

The thunderbolt glows when it’s charging

GREEN · GLOWING

Your system, at a glance

The illuminated ACECORE nameplate is a single, simple signal: is the unit on or off?

System On

The ACECORE panel glows steadily. Power is flowing and your PowerCell is actively working for your home.

- Status light active and readable
- App shows your system as "Connected"
- Backup load and solar charging both available

System Off

The ACECORE panel is unlit. No power is being drawn or supplied safe to inspect or disconnect.

- Status light dark
- App shows your system as "Offline"
- Safe moment for cleaning or a visual check

NOTE: After you switch the unit ON, allow a few seconds for the panel to light up and the status light to settle before reading it.

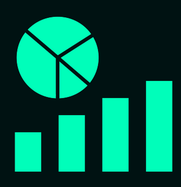
Control every appliance, individually

Each Smart Socket plugs between your appliance and the wall giving you a switch and a window into what it uses.



Remote On/Off

Turn any connected appliance on or off from the app, from anywhere.



Usage Insights

See exactly how much power each appliance draws, by the hour, day, or month.



Scheduling

Set appliances to switch on or off automatically at the times you choose.



Overload Protection

Each socket cuts power automatically if a connected appliance draws too much.



Smart Alerts

Get notified of unusual usage, so nothing runs longer or hotter than it should.

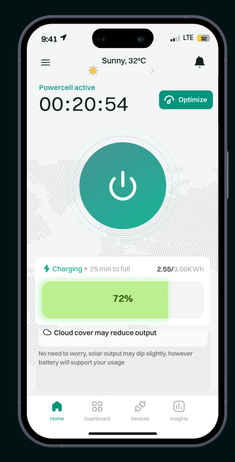


Whole-Home View

See every socket in your home on one screen, alongside your PowerCell's status.

Three sources. One seamless supply.

PowerCell always reaches for the cheapest, cleanest power first — switching sources automatically so your home never notices.



POWERCORE MOBILE APP

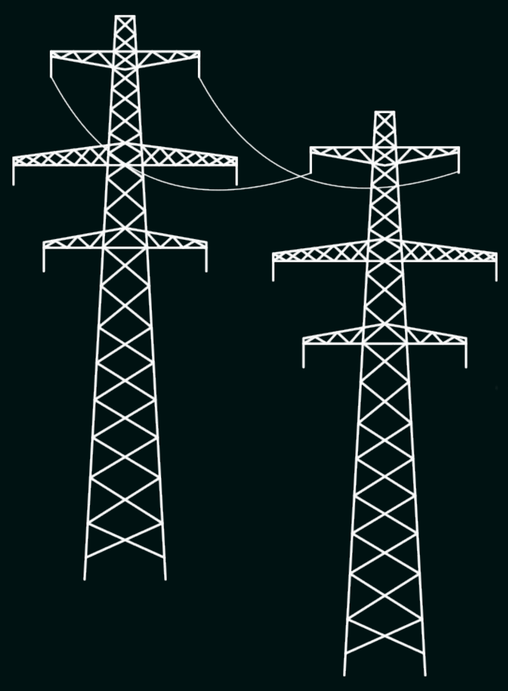
SUN

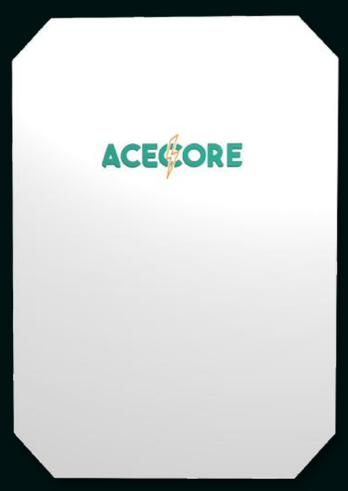


SOLAR PANEL



POWER GRID





POWERCELL



HOME

HOW IT DECIDES: Solar powers your home and charges your battery first. If solar runs short, PowerCell draws from the battery, then the grid. This happens in that order, without you lifting a finger.

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Part Two

Installation

Contains everything you need to about
the Powercell installation



Simple wiring.
Serious electricity.

PowerCell's installation is simple — but every connection carries real voltage. Treat it with full respect before you begin.

Qualified installer only
Every connection must be made by a qualified electrician or Acecore-approved installer.

Isolate before connecting
Confirm the solar array and the home distribution panel are both isolated before any cable is connected or removed.

Never in wet conditions
Do not install or connect cables in rain, standing water, or high humidity.

PV cables stay live in daylight
Solar panels generate voltage in sunlight even when disconnected from PowerCell. Treat the panel cable as live at all times during daylight.

Match the rated cable
Use only cable of the type and rating specified for PowerCell — never a substitute of lower rating.

Check before you power on
Inspect each connection for full insertion and correct polarity before switching anything on.

WARNING: Never connect or disconnect a cable while PowerCell is powered on. Always isolate first, connect or disconnect, then power on last.

From box to power

Installing the Powercell is very simple! No rocket science involved.

> ⚠ The PowerCell is a floor-mounted energy storage system and must be installed by an Acecore licensed installer. Do not attempt self-installation.

Connections

The system connects to three circuits via a 6-way terminal block inside the enclosure: AC output to your distribution board, AC input from the grid supply, and DC input from your solar array. Each AC line requires a 32Amp circuit breaker.

Software Integration

Once installed, press the System Control Button on the lower-right of the enclosure to power on the electronics. Open the PowerCell App and connect either via PowerCell Wi-Fi (offline) or your System ID and passcode (online). Switch the system ON from the app dashboard.

Smart Socket Setup

For Smart Socket setup, plug it in and enter its serial number under the Devices tab in the app.

For full wiring diagrams, terminal assignments, mounting dimensions, and step-by-step procedures, refer to the PowerCell Installation Sheet. If you have any challenge feel free to Contact us. See page 26 for our contact options



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Part Three

The Acecore App

Contains everything you need to know and properly use the Acecore App, so you can maximize and enjoy full control of your Powercell.



Set up the Acecore app in minutes

Your PowerCell works with or without the app, but with it comes additional monitoring, optimization, and remote control features.

01

Download the app

Search "Acecore" on the App Store or Google Play Store, and install it on your phone.

02

Create your account

To create an account, enter your Powercell ID and press “Continue” then enter Full Name, Email address,and a confirmed passcode across the following steps. Each field is followed by a Next or Continue Button to advance to next Screen.

03

Login

To Login, simply enter your Powercell ID and your Passcode and click on Continue.

04

Connect to Wi-Fi

Choose your home Wi-Fi network from the list and enter the password when prompted.

05

Start monitoring

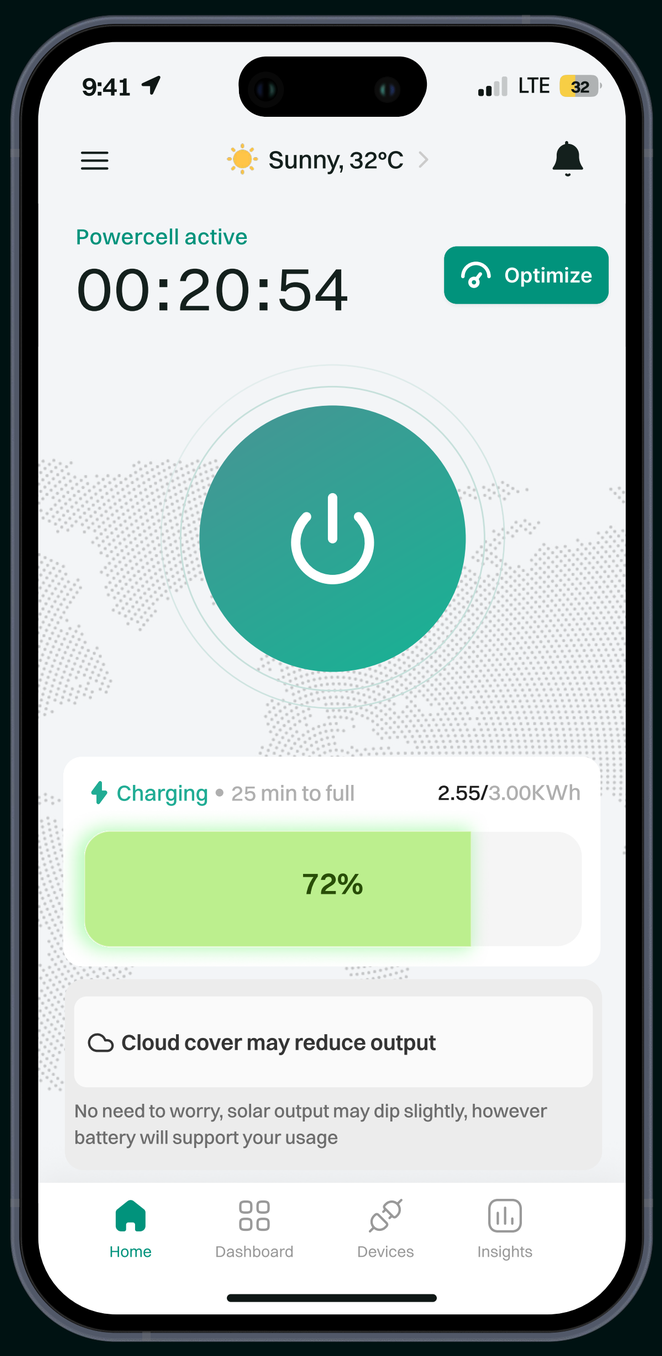
Once connected, your dashboard populates within a minute (see page 12 for a full tour).


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Your dashboard, explained

One screen, your whole system. Here's what each part of the home screen is telling you.



01 Active TIme

This time counter tells how long your system has been ON and running.

02 Optimize Button

This Button helps toggle on the Optimization features

03 Switch Button

This button turns the system ON/OFF

04 Battery Bar

This tells the battery percentage and charging status of the system.

Insights that pay for themselves

Beyond live numbers, the app studies your usage over time and turns it into guidance you can act on.

Usage Trends
See how your consumption, solar generation, and charging shift by day, week, or month.

Smart Recommendations
Get suggestions on when to run heavy appliances, based on your solar and tariff patterns.

Peak-Time Alerts
Know when you're about to draw from the grid at its most expensive, before it happens.

Historical Reports
Download a summary of your energy use and savings for any past period.

Battery Health
Track how your battery is ageing over time, and get a heads-up before performance dips.

Whole-Home Breakdown
See which appliances or sockets are driving your usage the most.

Your home, from wherever you are

Control of the systems works over the internet, no need to be on your home Wi-Fi. Changes you make reach your PowerCell within seconds from whereever you are.

Turn Sockets On/Off
Switch any connected appliance on or off, from across town or across the country.

Schedule Automations
Set sockets or charging behaviour to change automatically at times you choose.

Multiple Users
Invite family members to the same PowerCell, each with their own login.

Adjust Charging Priority
Change whether solar or grid charging is prioritised, right from the app.

Instant Alerts
Get a push notification the moment something needs your attention.

Works Anywhere
Control your system over Wi-Fi or mobile data. No VPN or special setup required.

NOTE : Remote control needs your PowerCell to have an active internet connection. If it's offline, changes will apply as soon as it reconnects.

Common problems, solved

Most issues clear up with one of these. If yours doesn't, turn to page 19 for support.

APP

App won't load

Close and reopen the app. If it persists, update to the latest version from your app store, or restart your phone.

CONNECTIVITY

PowerCell shows Offline in the app

Check your home Wi-Fi is working. If it is, restart your PowerCell's Wi-Fi module by toggling it off and on from the device menu.

CONNECTIVITY

Device won't connect during setup

Confirm you're using a 2.4GHz Wi-Fi network, most home routers also broadcast a 5GHz network the unit can't connect to.

DATA

Dashboard shows no data or old numbers

Pull down on the dashboard to refresh. If numbers are still stuck, check your device is online or your internet connection.

SMART SOCKET

A socket isn't responding

Confirm the socket has power at the wall, and that it still appears in your device list. Try removing and re-adding it.

ACCOUNT

Didn't receive a verification code

Check your spam folder, confirm your number or email is correct, and request a new code after 60 seconds.

Managing your account and device

The two things people ask about most: getting back into your account, and getting your PowerCell talking to it.

Account

- Forgot password? Tap “Forgot Password” on the login screen to reset it by email or SMS.
- Change your email or phone?
  Go to Profile > Account Details.
- Delete your account: Request this from Profile > Account Details > Delete Account.
- One account, multiple PowerCells: Add each additional unit from Profile > My Devices.

Device

- Re-pairing after a router change: Go to My Devices, select your PowerCell, and choose “Reconnect Wi-Fi.”
- Adding a device: Go to Devices, tap the “+” in the top right, and scan.
- Forgetting a device: Go to Devices and select “Forget Device.”
- Firmware updates: This applies automatically; you'll see a brief notice in the app when one completes.

Frequently asked questions

The questions we hear most from PowerCell owners.

Q. Does PowerCell work if my internet is down?

Yes. Your PowerCell keeps charging, discharging, and protecting your battery on its own, you'll just lose remote monitoring until you're back online.

Q. Can I use the app on more than one phone?

Yes. Log in with the same account on any number of phones, all changes and data stay in sync across all of them.

Q. How many Smart Sockets can I connect?

As many as your home needs. Each one appears individually in the app once paired.

Q. Will I lose settings during a power outage?

No. Your configuration is stored on the device itself and is not affected by outages.

Q. Can I control PowerCell without the app?

Yes, the unit runs automatically on its own logic. The app adds visibility and remote control, but isn't required for normal operation.

How we handle your data

Your PowerCell requires energy data to run features like insights and remote control. Here's what that involves.

Encrypted in Transit
Data between your PowerCell, the app, and Acecore's servers is encrypted end to end.

Only Energy Data
We collect system performance and usage data, never the contents of anything connected through a socket.

You Control Access
Only accounts you invite can view or control your PowerCell. Remove access at any time.

Secure Login
Enable two-factor authentication from Profile > Security for an extra layer of protection.

Your Data, On Request
Request a copy or deletion of your account data at any time. (see Contact Support).

Local-First Operation
Your PowerCell keeps working on its own logic even when disconnected, the cloud only adds visibility.

Talk to a real person

Have your PowerCell serial number ready, you'll find it on the rear nameplate, this is for a faster resolution.

EMAIL

info@acecore.tech

Best for detailed issues, replies within 1 business day.

PHONE

+234 706 757 9254

For urgent faults, available during business hours.

WHATSAPP

+234 901 529 4724

Message us directly for quick questions.

SUPPORT FORM

acecore.tech/support

You can also drop complaints or remarks on our website

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Part Four

Living With The Powercell

Contains everything you need to know about the daily usage of the Powercell and Acecore App



A few minutes, twice a year

PowerCell is built to run untouched. This is the entire maintenance routine.



Clear the vents

Wipe dust from the top and base vents with a dry cloth every few months, weekly in dusty areas.



Watch for alerts

Check the app occasionally for any fault notice, and act on it promptly.



Keep it dry

Never hose down, wet-wipe, or expose the unit to rain or splashing.



Don't block it

Keep the clearances from page 9 free for airflow at all times.



Leave it sealed

Never open the casing yourself. All internal service is for qualified technicians only.



Call it in

For anything beyond wiping the outside, contact Acecore support, see page 26.

Safety, once it's running

Installation safety is covered on page 8. These are the essentials for living with a working PowerCell.

No disassembly
Never open or attempt to repair the unit yourself. Contact a qualified technician.

Away from flame
Keep clear of open flame, and don't store flammable material nearby.

Away from water
Protect from rain, flooding, and high humidity at all times.

Never a frozen battery
Do not charge or operate the unit if it has been exposed to freezing temperatures.

Stop use if damaged
If the casing, cabling, or connectors show any damage, power down and contact support immediately.

No modifications
Never alter the unit, its cables, or its internal settings beyond what the app allows.

WARNING: If you notice burning smells, sparking, or unusual heat from PowerCell, power it down immediately and contact support.

Specifications

The figures worth knowing. Full technical documentation is available on request.

POWERCELL — SPEC PLATE	ACE-PWC-01
BUILD	
DIMENSIONS (H×W×D)	940 × 610 × 142 mm
MOUNTING	Floor-standing
ON-SITE WIRING	Inverter output, inverter input, and PV
BATTERY	
CHEMISTRY	$LiFePO_4$ (Lithium Iron Phosphate)
CONFIGURATION	16-cell series pack
PROTECTION	Overvoltage, overcurrent, over-temperature
POWER	
NOMINAL SYSTEM VOLTAGE	48 V
SOLAR CHARGE CURRENT	50 A, MPPT tracking
AC OUTPUT	Pure sine wave
CHARGING SOURCES	Solar or grid
CONNECTIVITY	
MONITORING	Wi-Fi, via the Acecore app
SMART SOCKET	Wireless, per-appliance control

Mechanical Specifications



Dimensions (H x W x T)	940 × 610 × 142 mm (62.8 × 29.7 × 6.3 in)
Total Weight	106 kg (233.69 lb)
Battery Assembly	79.5 kg (175.267 lb)
Components	26.5 kg (58.422 lb)
Mounting Options	Floor

We're here if you need us

This guide covers the essentials. For anything else, page 26 has every way to reach us.

WARRANTY

Standard coverage included

Every PowerCell is covered against manufacturing defects for the period stated on your sales documentation. Ask your installer for the full terms.

MANUFACTURER

Acecore Incorporations

PowerCell Division
Oyigbo, Rivers State, Nigeria

SUPPORT

See page 26

Email, phone, WhatsApp, or our online form. Pick whichever suits you.

THIS GUIDE

CORE USER GUIDE 1.0

Acecore may update this guide as PowerCell evolves. Ask your installer if you'd like the latest edition.

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THANK YOU

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Powercell Prime

POWER THAT UNDERSTANDS YOU